

# PACKMAR RION LOUJI

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## OBJECTIVE

Aspiring to pursue a master's degree in Computer Science with a concentration in Artificial Intelligence to gain expertise in medical AI, machine learning, and research methodology. An academic background in Linear Algebra, Partial Differential Equations, Probability and Random Processes, Signal Processing, and Robotics and Automation has provided a strong foundation for AI. Exploration of its real-world potential has been conducted through self-initiated work in healthcare and financial domains. Post graduation, the goal is to contribute to human-centric healthcare innovations, where intelligent systems significantly support clinical expertise and transform patient experiences.

## EDUCATION

**Bachelor of Engineering** – Electronics and Communication Engineering  
Loyola-ICAM College of Engineering and Technology, Anna University, Chennai

**July 2019 – June 2023**

**CGPA:** 8.66 / 10

**RELEVANT COURSES:** *Probability and Random Processes | Linear Algebra and Partial Differential Equations | Discrete-Time Signal Processing | Robotics and Automation*

## PROJECTS

### Pneumonia Detection using Image Processing [LICET]

**Dec 2022 – May 2023**

**Team Size:** 3

**Role:** Developer – Model Development & Deployment

#### Summary:

As part of a healthcare-focused initiative, we developed a deep learning model to detect pneumonia from chest X-ray images with an accuracy of 98.4%. The model was integrated into a mobile application, enabling real-time image uploads and instant predictions. This end-to-end system aimed to improve accessibility to diagnostic tools and support early detection, especially in remote or resource-limited settings.

### AI-Based Stock Analysis and Trend Prediction Tool [Independent]

**Aug 2022**

**Team Size:** 1

**Role:** Developer – Design & Implementation

#### Summary:

I developed an AI-based tool that fetches live stock data in real time and performs automated financial analysis to assess market conditions. By applying technical indicators and trend detection logic, the system generates immediate predictions to support quick investment decisions. This solution effectively reduced the time and effort involved in manual stock evaluation by automating both data retrieval and decision-making processes.

### Home Automation with Raspberry Pi [Hackathon]

**Oct 2022 – Nov 2022**

**Team Size:** 4

**Role:** Developer

#### Summary:

Developed a system to remotely control home appliances via a mobile app using Raspberry Pi and a Relay module. Won first prize in Pals Nexgen 3d Hackathon for this project, receiving a prize reward.

### Line Follower Robot [Hackathon]

**Sep 2021**

**Team Size:** 3

**Role:** Developer

#### Summary:

Designed a robot that autonomously follows path lines, which can be used in smart delivery, warehouse transport, and automated cleaning systems, reducing manual effort. Won second prize in this Hackathon.

## PROFESSIONAL EXPERIENCE

### Associate Software Developer, Pando (Quaking Aspen Technologies Pvt. Ltd), Chennai

**Feb 2024 – Present**

**Tech Stacks:** NodeJs, VueJs, Postgres, ExpressJs, FastifyJS, MongoDB, AWS, Jenkins, Sequelize, Redis

- Built scalable applications using Node.js, Vue.js, PostgreSQL, and related technologies and conducted thorough code reviews to uphold code quality and maintainability.
- Built a real-time logistics ranking system to streamline the evaluation of over 1,000+ transporter quotes across 2,000+ lanes during competitive auctions. To handle this scale efficiently, designed a multi-step architecture where each incoming or updated quote automatically triggered a chain of background tasks. Implemented Redis locks along with Bull Job Queues to manage concurrency and consistency, which eliminated deadlocks, reduced database load by 25%, and improved system responsiveness and reliability in high-pressure auction scenarios.
- Took complete ownership of a dedicated module, overseeing it end-to-end from development to deployment, and ensured the timely delivery of critical bug fixes and production issues.

- Led code revamps, handled project migrations, and addressed key security vulnerabilities such as SQL injection.
- Recognized with a team-level performance award for consistent adherence to clean and efficient coding practices.

## INTERNSHIPS

### Software Development Intern, Pando (Quaking Aspen Technologies Pvt. Ltd), Chennai

Oct 2023 - Feb 2024

**Skills:** NodeJs, VueJs, Postgres, ExpressJs, Sequelize, Redis

- Initially worked on a personal project, "Wallet", using PostgreSQL, Node.js, Vue.js, Sequelize, and Excel libraries to implement features like money transfer between accounts, balance viewing, and bulk uploads.
- Gradually started contributing to smaller Node.js tasks in a dedicated module, which helped in building hands-on experience and paved the way for larger responsibilities in professional projects.

### Trainee Auxiliary, BNP Paribas India Solutions Pvt. Ltd, Chennai

Jan 2023 – June 2023

**Skills:** Vermilion Reporting Suite, Java, Apache Drill, MapRDB

- Generated investment reports using Vermilion Reporting Suite with data sourced from Apache Drill and MapR-DB, and performed static code analysis on Java projects using SonarQube to enhance code quality.
- Gained exposure to key financial concepts, including mutual funds and stock market instruments, and deepened understanding of investment reporting and financial data analysis.

### Trainee Auxiliary, Defence Research and Development Organisation (DRDO), LRDE, Bangalore

July 2022 – Sept 2022

**Skills:** Matlab, Bistatic Radar Systems

- Explored and analyzed various RADAR systems, focusing on the positioning of transmitting and receiving antennas, and performed mathematical computations for beamforming, multibeam selection, and beam efficiency using 3 dB and 6 dB drops.
- Enhanced problem-solving skills in real-world defense applications, calculated coverage area and range cell values, and simulated results using MATLAB, and gained insights into signal optimization across different azimuth and elevation angles.

## TECHNICAL SKILLS

|                                    |                                                                        |
|------------------------------------|------------------------------------------------------------------------|
| <b>Languages:</b>                  | JavaScript, Java, MATLAB, Python                                       |
| <b>Frameworks &amp; Libraries:</b> | Node.js, Express.js, Fastify.js, Vue.js, Pinia, Sequelize, Mocha, Chai |
| <b>Databases:</b>                  | PostgreSQL, MongoDB                                                    |
| <b>DevOps &amp; CI/CD Tools:</b>   | AWS, Jenkins, Docker, Git, Redis                                       |
| <b>AI Tools:</b>                   | ChatGPT, GitHub Copilot, Amazon Q                                      |
| <b>Editors &amp; IDEs:</b>         | VS Code, WebStorm, Jupyter Notebook, Cursor, Eclipse, Overleaf         |
| <b>Others:</b>                     | REST APIs, Agile                                                       |

## CERTIFICATIONS / COURSES

1. **The Enterprise Edge (Learning Program)** - Pando - June 2025
2. **Machine Learning** - Perfect e-learning - August 2023
3. **Project-Based Learning Team2Solve & Code2Build** - ICT Academy - April 2023
4. **Certificate Studies in Sound Engineering and Production** - Risaya Academy - Jan 2023
5. **Uninhabited Aircraft System (UAS)** - SAE India (Aerothon) - June 2022
6. **AWS Cloud Architecting** - ICT Academy - September 2021
7. **Machine Learning Internship Program** - Skill Safari - July 2021
8. **SQL for Business Analytics** - Turnip Innovations Pvt. Ltd - April 2021

## ACTIVITIES AND ACHIEVEMENTS

- **Trailblazer Award, Pando, Chennai, June 2025**  
Honored as a high-impact performer for consistently delivering clean, reliable code, actively identifying and resolving bugs, and contributing to overall quality through proactive code reviews and issue resolution.
- Won the first prize in Pals Nexgen 3d Hackathon, an initiative by pan-IIT Alumni, October 28, 2022 - November 6, 2022
- Won the second prize in Pals Nexgen 3d Hackathon, an initiative by pan-IIT Alumni, September 1, 2021 - September 9, 2021
- Represented the school and College as a member of the football team, LICET, 2019
- Performed as a musician in the department band, LICET, Enginea, October 2023